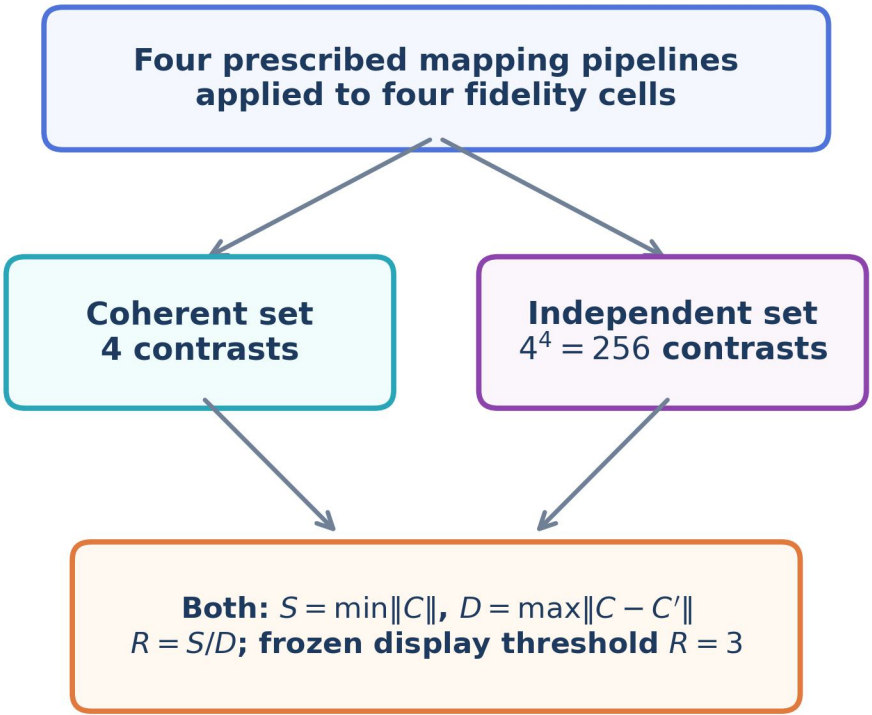


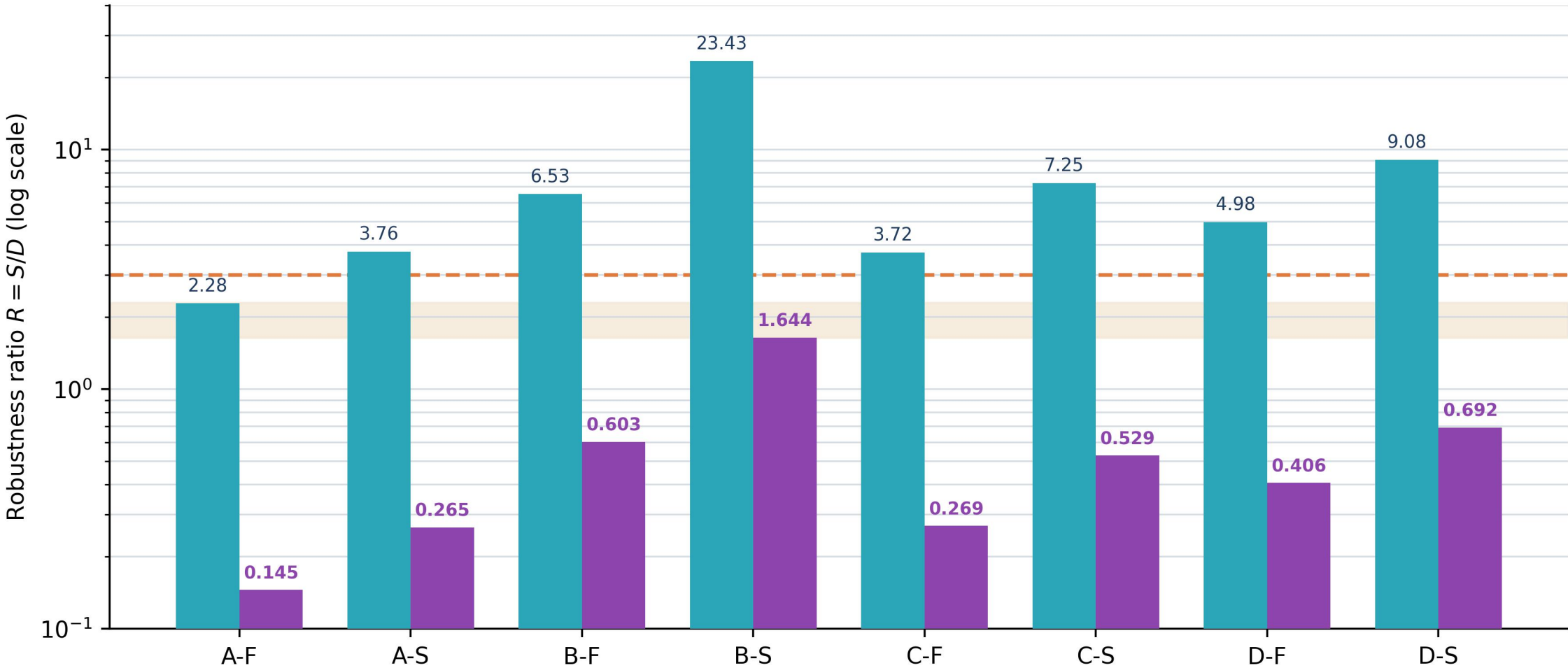
Dependence construction changes real-CFD mapping-robustness classification

All-eight reversal interval Coherent 4-pipeline set
Frozen threshold $R = 3$ Independent 4^4 recombinations

(a) Same outputs and set functionals



(b) All objectively included complete lattices (4 geometries $\times$ 2 supports)



Frozen $R=3$: 7/8 paired reversals; independent passes: 0/8. All eight reverse for $1.6443 < \text{threshold} < 2.2832$. Exact-diameter expansion: $10.83\text{--}15.71\times$.

A: alph05-10071-2024; B: alph15-10929-3036; C: alph15-13929-2024; D: alph15-7929-2024. F: fine F100 support; S: shared F000 support.